

General Chemistry (CHEM 1310)

Midterm Examination 1

Name	GT ID (90...)
Section (A or B)	

Indicate your response to each question in the box or on the line(s) provided. For multiple-choice questions, fill in the circle next to your choice clearly and completely. For multiple-answer questions, fill in the circles completely next to statements you wish to select.

Academic Integrity Guidelines

1. This quiz is closed note, closed book, and closed Internet. You are permitted only a single crib sheet and a calculator lacking Internet connectivity.
2. You may not access any other resource or website aside from this document for the duration of the exam.
3. You may not collaborate with any other person (in person or electronically) on any portion of the exam, including general discussion of exam question topics prior to exam submission by all parties.
4. You are not permitted to screen-capture or otherwise record the questions on this exam.
5. All work submitted for this exam must be your own.

I affirm that I have abided by all academic integrity guidelines as listed above.

Signature _____

Do not open this exam until you are instructed to do so.

1. **6 points.** Which of the following is the correct symbol for an atom or ion containing 45 protons, 58 neutrons, and 42 electrons?

A. ☐ $^{97}\text{Rh}^{3+}$
B. ☐ $^{58}\text{Rh}^{3+}$
C. ☐ $^{103}\text{Rh}^{3+}$
D. ☐ $^{103}\text{Rh}^{3-}$
E. ☐ $^{97}\text{Rh}^{3-}$

2. **6 points.** Enter the correct element symbol (e.g., "He") based on each of the following descriptions.

The noble gas in the fifth period _____

The transition metal with atomic number 44 _____

The lanthanide with the smallest atomic number _____

The alkaline earth metal in the third period _____

3. **5 points.** A hypothetical element has only two stable isotopes, A and B. The atomic masses of the isotopes are listed below. What is the average atomic mass in atomic mass units (amu) of the hypothetical element if 71,222 of every 100,000 atoms are atoms of isotope A? Enter your response to the nearest 0.001 amu.

Isotope A = 49.00 amu

Isotope B = 47.00 amu

amu

4. **6 points.** The compound retinol has the formula $\text{C}_{20}\text{H}_{30}\text{O}$. Indicate whether each statement below regarding this compound is true or false.

[☐ True ☐ False] The molar mass of this compound is 286.5 g/mol.

[☐ True ☐ False] Each molecule of retinol in a bulk sample has an identical mass of 286.5 g/mol.

[☐ True ☐ False] Although the average number of carbon atoms per molecule is 20, retinol molecules contain a distribution of different numbers of carbon atoms.

[☐ True ☐ False] In a sample of retinol weighing 286.5 g, there are 6.022×10^{23} oxygen atoms.

5. **6 points.** Indicate whether each of the following chemical names corresponds correctly to the given formula.

S_4N_4 | tetrasulfur tetranitride [☐ Correct ☐ Incorrect]

SO_3 | sulfur(IV) oxide [☐ Correct ☐ Incorrect]

$CaSO_3$ | calcium sulfate [☐ Correct ☐ Incorrect]

$Fe_3(PO_4)_2$ | iron(II) phosphate [☐ Correct ☐ Incorrect]

6. **6 points.** How many moles of **sodium** are present in a sample of Na_2CrO_4 weighing 310.2 g? The molar mass of Na_2CrO_4 is 161.97 g/mol. Enter your response in moles (mol) to the nearest 0.01 mol.

mol

7. **6 points.** In a compound containing only nitrogen and oxygen, the percent by mass of nitrogen is 25.94%. What is the empirical formula of this compound?

- A. ☐ N_2O
B. ☐ N_2O_5
C. ☐ NO_2
D. ☐ NO
E. ☐ N_2O_3

8. **6 points.** Consider what happens when aqueous calcium nitrate and aqueous sodium carbonate are mixed. Using the dropdown menus below, indicate the identity of the solid product that forms (if any) and the molar ratio of calcium nitrate to sodium carbonate in the balanced chemical equation.

Solid product: [☐ Sodium nitrate ☐ Calcium carbonate ☐ None]

Calcium nitrate/sodium carbonate ratio: [☐ 1:1 ☐ 1:2 ☐ 2:1 ☐ 2:3 ☐ 3:2]

9. **6 points.** Indicate whether each solute below forms a strong electrolyte, weak electrolyte, or nonelectrolyte solution when dissolved in water.

Acetic acid (HO_2CCH_3)

[☐ strong electrolyte ☐ weak electrolyte ☐ nonelectrolyte]

1-Propanol ($\text{C}_3\text{H}_7\text{OH}$)

[☐ strong electrolyte ☐ weak electrolyte ☐ nonelectrolyte]

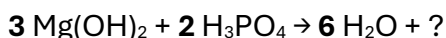
Potassium perchlorate (KClO_4)

[☐ strong electrolyte ☐ weak electrolyte ☐ nonelectrolyte]

10. **5 points.** What mass of calcium nitrate ($\text{Ca}(\text{NO}_3)_2$, 164.09 g/mol) is needed to prepare 4.25 L of a solution that is 0.275 M in **nitrate ions**? Enter your response in grams (g) to the nearest 0.1 g.

g

11. **5 points.** In an experiment involving the combination of magnesium hydroxide and phosphoric acid, the stoichiometry in bold below was observed. What is the chemical formula of the missing product?

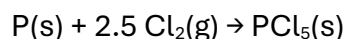
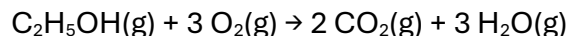
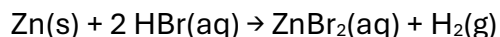


- A. ☐ $\text{Mg}_2(\text{PO}_4)_3$
B. ☐ MgPO_4
C. ☐ $\text{Mg}_3(\text{PO}_4)_2$
D. ☐ Mg_3PO_4
E. ☐ $\text{Mg}(\text{PO}_4)_2$

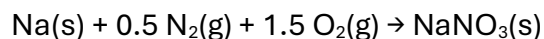
12. **5 points.** What aqueous species are produced, and in what amounts, when 1 mole of NaNO_3 is dissolved in water?

- A. ☐ 1 mol Na^+ , 1 mol $\text{N}^{5+}(\text{aq})$, and 3 mol $\text{O}^{2-}(\text{aq})$
B. ☐ 1 mol $\text{Na}^+(\text{aq})$, 0.5 mol $\text{N}_2(\text{aq})$, and 1.5 mol $\text{O}_2(\text{aq})$
C. ☐ 1 mol $\text{Na}(\text{aq})$ and 1 mol $\text{NO}_3(\text{aq})$
D. ☐ 1 mol $\text{NaNO}_3(\text{aq})$; this compound does not dissociate in solution.
E. ☐ 1 mol $\text{Na}^+(\text{aq})$ and 1 mol $\text{NO}_3^-(\text{aq})$

13. **6 points.** Classify each of the following reactions as synthesis, decomposition, single replacement, double replacement, or combustion.

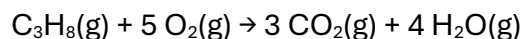


14. **5 points.** How many moles of oxygen gas are needed to prepare 8.40 moles of sodium nitrate according to the balanced chemical equation below? Assume excess amounts of all other reactants.

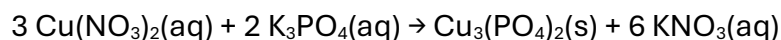


- A. ☐ 8.40 mol
B. ☐ 5.60 mol
C. ☐ 4.20 mol
D. ☐ 12.6 mol
E. ☐ 16.8 mol

15. **5 points.** What mass of carbon dioxide (CO_2 , 44.0 g/mol) is produced when 10.5 kg of propane (C_3H_8 , 44.1 g/mol) is fully combusted according to the balanced reaction below? Assume that the reaction goes to completion. Enter your response in kilograms (kg) to the nearest 0.1 kg.

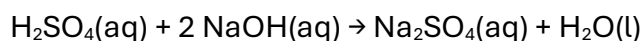
 kg

16. **5 points.** Masses of 0.187 g **each** of $\text{Cu}(\text{NO}_3)_2$ (187.55 g/mol) and K_3PO_4 (212.27 g/mol) are mixed in aqueous solution and react according to the balanced chemical equation below. Solid $\text{Cu}_3(\text{PO}_4)_2$ (380.58 g/mol) with a mass of 0.095 g was obtained. What percent yield is implied by this result?



- A. ☐ 63%
B. ☐ 101%
C. ☐ 56%
D. ☐ 75%
E. ☐ 86%

17. **5 points.** An aqueous solution of sulfuric acid with volume 250. mL and molarity 0.125 M was subjected to complete neutralization according to the balanced chemical equation below, using 300. mL of a sodium hydroxide solution. What is the minimum molarity of the sodium hydroxide solution required to accomplish this? Enter your response in moles per liter (mol/L, M) to the nearest 0.001 M.



M

18. **6 points.** P_4 , known as “white phosphorus,” is a polyatomic form of phosphorus. Indicate whether each statement regarding this substance is true or false.

[☐ True ☐ False] The average mass of a molecule of P_4 is 30.97 amu.

[☐ True ☐ False] The empirical formula of white phosphorus is P.

[☐ True ☐ False] There are four times as many phosphorus atoms as P_4 molecules in any sample of white phosphorus.

[☐ True ☐ False] P_4 is best described as a compound because it contains molecules rather than atoms.